

1. Given grammar $G = (\{S, H\}, \{a, c, d, e\}, \{S \rightarrow a^2 S e \mid H, H \rightarrow c H d^2 \mid cd\}, S)$, find the generated language (+proof). - 25P

$$(S) \xRightarrow{(2)} H \xRightarrow{(4)} cd$$

$$(3) \xRightarrow{(3)} c H d^2 \xRightarrow{(4)} c^2 d^4$$

$$(3) \xRightarrow{(3)} c^3 H d^6 \xRightarrow{(4)} c^4 d^8$$

$$(3) \xRightarrow{(3)} c^4 H d^8 \xRightarrow{(4)} c^5 d^{10}$$

$$(3) \xRightarrow{(3)} \dots$$

$$? \quad H \xRightarrow{*} \underbrace{c^n d^{2n-1}}_{P(n)}, \forall n \in \mathbb{N}^*$$

I ? $P(1)$ is true

$$H \xRightarrow{(4)} cd = c^1 d^{2 \cdot 1 - 1} \Rightarrow P(1) \text{ is true}$$

II $P(k) \xRightarrow{?} P(k+1), \forall k \in \mathbb{N}^*$

$$P(k) \text{ is true} \Rightarrow H \xRightarrow{*} c^k d^{2k-1} \xRightarrow{(3)}$$

$$\Rightarrow H \xRightarrow{*} c c^k d^{2k-1} d^2 = c^{k+1} d^{2(k+1)-1} \Rightarrow P(k+1) \text{ is true}$$

$$I, II \Rightarrow P(n) \text{ is true, } \forall n \in \mathbb{N}^* \Rightarrow$$

$$\Rightarrow H \stackrel{*}{\Rightarrow} c^m d^{2m-1}, \forall m \in \mathbb{N}^*$$

$$S \stackrel{(2)}{\Rightarrow} H$$

$$\stackrel{(1)}{\Downarrow} b^2 S e \stackrel{(2)}{\Rightarrow} b^2 H e$$

$$\stackrel{(1)}{\Downarrow} b^4 S e^2 \stackrel{(2)}{\Rightarrow} b^4 H e^2$$

$$\stackrel{(1)}{\Downarrow} b^6 S e^4 \stackrel{(2)}{\Rightarrow} b^6 H e^4$$

$$\stackrel{(1)}{\Downarrow} b^8 S e^6 \dots$$

$$\begin{array}{c} ? S \stackrel{*}{\Rightarrow} b^{2m} H e^m, \dots, \forall m \in \mathbb{N} \\ \hline P(m) \end{array}$$

$$\begin{array}{c} ? S \stackrel{*}{\Rightarrow} b^{2m} c^m d^{2m-1} e^m, \forall m \in \mathbb{N}, \forall n \in \mathbb{N}^* \\ \hline P(m, n) \end{array}$$

$$I ? P(1, 0) \text{ is true } \Leftrightarrow S \stackrel{*}{\Rightarrow} cd$$

$$S \stackrel{(2)}{\Rightarrow} H \stackrel{(4)}{\Rightarrow} cd \Rightarrow P(1, 0) \text{ is true}$$

$$k > n, l > 0$$

$$k \in \mathbb{N}^*, l \in \mathbb{N}$$

$$\text{II } P(k, l) \stackrel{?}{\Rightarrow} P(k+1, l), P(k, l+1), P(k+1, l+1)$$

$$P(k, l) \text{ is true} \Rightarrow S^* \Rightarrow b^{2l} c^k d^{2k-1} e^l$$

$$\begin{array}{c} \xRightarrow{*} \\ \text{ind. hyp.} \end{array} \xRightarrow{(3)} \xRightarrow{*} \xRightarrow{(2)}$$

$$H \xRightarrow{(4)} cd \xRightarrow{(3)} c^{k+1} d^{2(k+1)-1} \xRightarrow{(2)} S^* \Rightarrow c^{k+1} d^{2(k+1)}$$

$$\xRightarrow{(1)} S^* \Rightarrow b^{2l} c^{k+1} d^{2(k+1)-1} \Rightarrow P(k+1, l) \text{ is true}$$

$$H \xRightarrow{(4)} (d \xRightarrow{(3)} c^k d^{2k-1}) \xRightarrow{(2)} S^* \Rightarrow c^k d^{2k+1}$$

$$\xRightarrow{(1)} S^* \Rightarrow b^{2(l+1)} c^k d^{2k+1} e^l \Rightarrow P(k, l+1) \text{ is true}$$

by applying the same logic, we can deduce that

$$P(k, l) \Rightarrow P(k+1, l+n)$$

$$\Rightarrow \underbrace{\{ b^{2m} c^m d^{2m-1} e^m \mid m \in \mathbb{N}^*, m \in \mathbb{N} \}}_L \subseteq L(G)$$

All grammar productions have been used in all possible combinations $\Rightarrow$ valid justif.

$$L \subseteq L(G), L(G) \subseteq L \Rightarrow L(G) = L$$

2. Find grammars that generate the following languages

A. $L_1 = \{ x^n y^n \mid n \in \mathbb{N} \} + \text{proof}$

It is enough to find G_1 s.t. $L_1 \subseteq L(G_1)$ and $L(G_1) \subseteq L_1$

Set $G_1 = (N, \Sigma, P, S)$

$$N = \{ S \}$$

$$\Sigma = \{ x, y \}$$

$$P: S \xrightarrow{(1)} x S y \mid \xrightarrow{(2)} \epsilon$$

I $L_1 \stackrel{?}{\subseteq} L(G_1)$

$$L_1 \subseteq L(G_1) (\Leftrightarrow) \forall n \in \mathbb{N}: x^n y^n \in L(G_1)$$

Get $n \in \mathbb{N}$ - fixed

$$S \xRightarrow{(2)} \epsilon = x^0 y^0 \xRightarrow{(1)} x y \xRightarrow{(1)} x^2 y^2 \Rightarrow$$

$$\Rightarrow S^* \Rightarrow x^n y^n \Rightarrow x^n y^n \in L(G_1) \Rightarrow L_1 \subseteq L(G_1)$$

$L_1 = \{ x^n y^n \mid n \in \mathbb{N} \}$

$$\text{II } L(G_1) \stackrel{?}{\subseteq} L_1 \Leftrightarrow ? S \text{ generates } x^n y^n, n \in \mathbb{N}$$

$$S \stackrel{(2)}{\Rightarrow} \varepsilon = x^0 y^0 \in L_1$$

$$\stackrel{(1)}{\Downarrow} xSy \stackrel{(2)}{\Rightarrow} xy \in L_1$$

$$\stackrel{(1)}{\Downarrow} x^2Sy^2 \stackrel{(2)}{\Rightarrow} x^2y^2 \in L_1$$

$$\stackrel{(1)}{\Downarrow} x^3Sy^3 \stackrel{(2)}{\Rightarrow} x^3y^3 \in L_1$$

$\Downarrow \dots$

$\Rightarrow S$ only generates $x^n y^n, n \in \mathbb{N}$

$$\Rightarrow L(G_1) \subseteq L_1$$

$$\text{I, II} \Rightarrow L(G_1) = L_1$$

$$\text{B. } L_2 = \{a^m b^{2m} \mid m \in \mathbb{N}^*\} + \text{proof}$$

$$\text{Let } G_2 = (N, \Sigma, P, S)$$

$$N = \{S\}$$

$$\Sigma = \{a, b\}$$

$$P: S \rightarrow aSb^2 \mid ab^2$$

$$\text{I } L_2 \stackrel{?}{\subseteq} L(G_2) \Leftrightarrow \forall m \in \mathbb{N}^*: a^m b^{2m} \in L(G_2)$$

Let $m \in \mathbb{N}^*$ - fixed

$$S \stackrel{(2)}{\Rightarrow} ab^2 \stackrel{(1)}{\Rightarrow} a^2 b^4 \stackrel{(1)}{\Rightarrow} \dots \stackrel{(1)}{\Rightarrow} a^{n-2} b^{2n}$$

$$S \stackrel{*}{\Rightarrow} a^m b^{2m} \Rightarrow a^m b^{2m} \in L(G_2) \Rightarrow L_2 \subseteq L(G_2)$$

$$\text{II } L(G_2) \stackrel{?}{=} L_2$$

$$S \Rightarrow_{(2)} a b^2 \in L_2$$

$$(1) \Downarrow a S b^2 \Rightarrow_{(2)} a^2 b^4 \in L_2$$

$$(1) \Downarrow a^2 S b^4 \Rightarrow_{(2)} a^3 b^6 \in L_2$$

$$(1) \Downarrow a^3 S b^6 \Rightarrow_{(2)} a^4 b^8 \in L_2$$

$\Rightarrow S$ only generates $a^n b^{2n}, \forall n \in \mathbb{N}^*$

$$\Rightarrow L(G_2) \subseteq L_2$$

$$\text{I, II} \Rightarrow L(G_2) = L_2$$

C. $L_3 = \{a^n b^m \mid n, m \in \mathbb{N}^*\}$ - regular grammar required + proof

$$\text{Let } G_3 = (N, \Sigma, P, S)$$

$$N = \{S, H\}$$

$$\Sigma = \{a, b\}$$

$$P: S \Rightarrow_{(1)} aS \mid_{(2)} aH$$

$$H \Rightarrow_{(3)} bH \mid_{(4)} b$$

right-linear

$$\text{I } L_3 \stackrel{?}{\subseteq} L(G_3) \Leftrightarrow \forall n, m \in \mathbb{N}^*: a^n b^m \in L(G_3)$$

Let $n, m \in \mathbb{N}^*$ fixed

$$H \Rightarrow_{(4)} b \xRightarrow_{(3)} b^{m-1} b^m \quad (i)$$

$$S \Rightarrow a H \xRightarrow[(1)]{n-1} a^n H \xRightarrow{(i)} a^n b^m \Rightarrow$$

$$\Rightarrow S^* \Rightarrow a^n b^m \Rightarrow a^n b^m \in L(G_3) \\ \Rightarrow L_3 \subseteq L(G_3)$$

$$\text{II } L(G_3) \stackrel{?}{\subseteq} L_3 \Leftrightarrow ? S \text{ only generates } a^n b^m, n, m \in \mathbb{N}^*$$

$$a) ? H \text{ only generates } b^m, m \in \mathbb{N}^*$$

$$H \Rightarrow b$$

$$\Downarrow (3) \quad b H \Rightarrow b^2$$

$$\Downarrow (3) \quad b^2 H \Rightarrow b^3$$

$$\Downarrow (3) \quad b^3 H \Rightarrow b^4$$

$$\Downarrow \dots$$

$$\Rightarrow H \text{ only generates } b^m, m \in \mathbb{N}^*$$

$$ii) ? S \text{ only generates } a^n H, n \in \mathbb{N}^*$$

$$S \Rightarrow a H$$

$$\Downarrow (1) \quad a S \Rightarrow a^2 H$$

$$\Downarrow (1) \quad a^2 S \Rightarrow a^3 H$$

$$\Downarrow (1) \quad a^3 S \Rightarrow a^4 H$$

$$\Rightarrow S \text{ only generates } a^n H, n \in \mathbb{N}^*$$

a), b), (2) $\Rightarrow G_3$ only generates $a^m b^m, m \in \mathbb{Z}$

(Considering that the other prod. rule of S besides (2) is (1), and that applying (1) still generates only results in L_3 , then this should be sound.) ~~I realized late that I could have had $N = \{S, A, B\}$, $\Sigma = \{a, b\}$, $P = \{S \rightarrow AB\}$ (not regular)~~

$$\Rightarrow L(G_3) \subseteq L_3$$

$$\text{I, II} \Rightarrow L(G_3) = L_3$$